Dr. Wending (William) Mai

Website: www.em-time.com
Email: wdm@ieee.org | Phone: (814) 857-9312

Computation Scientist | R&D Engineer | HPC & GPU Computing Specialist

PROFESSIONAL SUMMARY

Computational Electromagnetics specialist with 10+ years of experience in numerical EM methods, high-performance computing, and scientific software development. Deep expertise in DGTD, FEM, hp-adaptive meshing, GPU acceleration, and industry-grade solver development. Proven record of innovation across DARPA-funded research, commercial EDA tools, and peer-reviewed publications, including recognition for rewardable trade secrets.

TECHNICAL EXPERTISE

Computational & Numerical Methods	• Computational Electromagnetics (CEM), with emphasis on **Discontinuous Galerkin Time Domain (DGTD) and FEM** • **hp-adaptive mesh refinement (AMR)** and advanced error indicators • Wave–circuit co-simulation (MNA / SPICE) • 2.5D high-efficiency algorithms for PCB and EMC simulation
High-Performance Computing	• Multi-CPU and GPU parallelization (**OpenMP, OpenACC**) • Development and profiling on **Linux (NVIDIA HPC SDK)** and **Windows (Visual Studio)**
Programming & Tools	• **C++, Python, Fortran, MATLAB** • Machine learning with **scikit-learn** • HFSS, CST Microwave Studio, COMSOL (all with **script-based automation**)
Physics & Modeling	• Electromagnetic theory for **dispersive, nonlinear, and time-varying materials** • Knot electromagnetics combining **chirality and symmetry**

Professional Experience

Ansys. Inc, Canonsburg, PA, USA

Senior Research and Development Engineer 2022-2026

- Led enhancement of **hp-adaptive mesh refinement** in HFSS using **novel error indicators and machine-learning-assisted strategies**, enabling more efficient allocation of computational resources.
- Implemented into FEM techniques for large-scale EM analysis, achieving **significant reduction in degrees of freedom (DoF)** while preserving accuracy.
- Innovations in hp-AMR and error estimation recognized internally as a **"Rewardable Trade Secret."**
- Collaborated cross-functionally with Solver, ComEngine, UI, and Product teams.

Pennsylvania State University, State College, PA, USA

Postdoctoral Scholar 2019-2022

- Led a software development team of **3 postdoctoral scholars and 3 PhD students**.
- Design and implemented an inhouse DGTD solver featuring:
  - Efficient **direct solvers for nonlinear materials** without iterative schemes
  - Support for **time-varying permittivity/permeability**
  - Library of **thousands of dispersive materials** extracted from *refractiveindex.info* and fitted using generalized dispersion models
  - **Wave–circuit co-simulation** integrated with MNA/SPICE

- **Multi-CPU and GPU acceleration** via OpenMP and OpenACC
- Contributed to **two DARPA-funded electromagnetic research projects**.
- Integrated **machine learning** with EM simulation to develop:
  - Noble-metal dispersion models in the optical regime
  - Equivalent-circuit extraction for microwave antennas and circuits
- Co-developed preprocessing and post-processing GUI tools.

Visiting Scholar 2017-2019

- Developed DGTD solvers for **optical dispersive metamaterials** under a DARPA program.
- Proposed **chiral and rotationally symmetric knot structures** with novel polarization properties:
  - Linear-to-circular polarization conversion independent of excitation direction
  - 90° polarization rotation independent of incident polarization angle

University of Electromagnetic Science and Technology of China, Chengdu, China

Research Associate 2015-2019

- Proposed an efficient **2.5D DGTD method** for high-speed PCB EMC simulation
- Led microwave imaging development for **low-cost, fast stroke detection** using subspace optimization methods (SOM).
- Contributed to **EPSRC (UK)-funded projects** on microwave brain imaging for stroke detection.

Texas Instrument, Shanghai, China & Dallas, TX, USA

Engineer 2010-2012

Provided application engineering support, technical training, and chip customization in collaboration with R&D teams.

Lenovo, Chengdu, China

Student Researcher 2009

Conducted EMC and reliability design and testing for laptop platforms

EDUCATION

Doctor of Philosophy in Electromagnetic and Microwave Techniques: 2019

University of Electronic Science and Technology of China, Chengdu, China

Master of Science in Radio Physics: 2010

University of Electronic Science and Technology of China, Chengdu, China

Bachelor of Science in Information Science and Technology: 2007

University of Electronic Science and Technology of China, Chengdu, China

ACTIVITIES

Session Chair / Co-Chair, *Advances in Electromagnetics of Complex Media and Metamaterials*, ACES 2021

Author of 21 journal articles in IEEE, Optica, Physical Review, etc.

Reviewer of 50+ Journal papers

Senior Member of IEEE, **Life Member** of OSA(OPTICA) and ACES

More: www.em-time.com